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September 2024

A rehabilitation algorithm after lateral ankle sprains in professional football (soccer): An approach based on clinical practice guidelines. International Journal of Sports Physical Therapy, 19(7): 910-922, 2024.

Age of first exposure to contact sports is not associated with worse later-in-life brain health in a cohort of community dwelling older men. Clinical Journal of Sport Medicine, Ahead of Print: 1-8, 2024.

Antidoping testing and testicle cancer diagnosis: An unexpected helper. Journal of Science and Medicine in Sport, Ahead of Print: 1-3, 2024.

Artificial intelligence applications in the football codes: A systematic review. Journal of Sports Sciences, Ahead of Print: 1-17, 2024.

Association of genetic profile with muscle mass gain and muscle injury prevention in professional football (soccer) players after creatine supplementation. Nutrients, 16: 2511, 2024.

Athlete monitoring perspectives of sports coaches and support staff: A scoping review. International Journal of Sports Science & Coaching, Ahead of Print: 1-20, 2024.

Between-microcycle variability of external soccer training loads through the evaluation of a contemporary periodization training model 'CUPs'. International Journal of Sports Science & Coaching, Ahead of Print: 1-11, 2024.

Blood, Sweat, and Tears: Implications for hydration testing in combat sports – Investigating body mass loss and biomarker changes. Scandinavian Journal of Medicine & Science in Sports, 34:e14701, 2024.

Bye-Bye, Bye Advantage: Estimating the competitive impact of rest differential in the National Football League. arXiv:2408.10867, 2024.

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Can caffeine change the game? Effects of acute caffeine intake on specific performance in intermittent sports during competition: A systematic review and meta-analysis. International Journal of Sports Physiology and Performance, Ahead of Print: 1-17, 2024.

Cannabinoid Therapy in Athletics: A review of current cannabis research to evaluate potential real-world cannabinoid applications in sport. Sports Medicine, Ahead of Print: 1-27, 2024.

Classification of myo-connective tissue injuries for severity grading and return to play prediction: A scoping review. Journal of Science and Medicine in Sport, Ahead of Print: 1-40, 2024.

Comparing the maximal horizontal deceleration demands between a novel acceleration to deceleration assessment and the 505 change of direction test. Journal of Human Sport and Exercise, 19(3): 803-814, 2024.

Correlation of player and imaging characteristics with severity and missed time in National Football League professional athletes with hamstring strain injury – A retrospective review. American Journal of Sports Medicine, Ahead of Print: 1-9, 2024.

Does draft currency promote competitive balance? An empirical investigation of the National Football League 2002-2021. Journal of Sports Economics, Ahead of Print: 1-23, 2024.

Effect of data-processing methods on acceleration summary metrics of GNSS devices in elite Australian football. Sensors, 24: 4383, 2024.

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Predicting sprint performance from the vertical and horizontal jumps in National Football League combine athletes. Journal of Strength and Conditioning Research, 38(8): 1433-1439, 2024.

Preseason and in-season high-speed running demands of 2 professional Australian Rules Football teams. Sports Health, Ahead of Print: 1-12, 2024.

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What are the protocols and resources for sport-related concussion among top NCAA football programs? A cross-sectional survey of A5 schools. Clinical Journal of Sport Medicine, Ahead of Print: 1-7, 2024.

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A rehabilitation algorithm after lateral ankle sprains in professional football (soccer): An approach based on clinical practice guidelines. International Journal of Sports Physical Therapy, 19(7): 910-922, 2024.

Abstract - Lateral ankle sprain (LAS) is one of the most common types of injury in professional football (soccer) players with high risk of recurrence. The rehabilitation after LAS in professional football players is often still time-based and relies on anecdotal experience of clinicians. There is still a lack of utilization of criteria-based rehabilitation concepts after LAS in professional football. The aims of this clinical commentary are (1) to critically discuss the need for criteria-based rehabilitation concepts after LAS in professional football players, (2) to highlight the current lack of these approaches and (3) to present a novel clinical guideline-based rehabilitation algorithm. Short time-loss (15 days) and high recurrence rate (17%) raise the question of trivialization of LAS in professional football. Despite consequences for many stakeholders involved (players, teams, clubs, insurers), there is still a lack of of criteria-based, step-by-step approaches. The use of a criteria-based rehabilitation approach might reduce the high recurrence rate after LAS in professional football players and will lead, in turn, to increased long-term player availability. Practical experiences of the authors demonstrate the feasibility of such an approach. The effectiveness of this novel rehabilitation algorithm remains to be evaluated in future studies. Level of Evidence: 5.

Age of first exposure to contact sports is not associated with worse later-in-life brain health in a cohort of community dwelling older men. Clinical Journal of Sport Medicine, Ahead of Print: 1-8, 2024.

Objective: To evaluate whether early age of first exposure to contact sports (AFE-CS) is associated with worse long-term brain health outcomes.

Design: A cross-sectional, survey study of older men with a history of contact sport participation was completed.

Setting: Tertiary care facility.

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Participants: A cohort of community-dwelling older men dichotomized by using AFE-CS (<12 years vs ≥12 years).

Interventions: Independent variables included a dichotomized group of AFE-CS (<12 years vs ≥12 years).

Main outcome measures: Brain health outcomes measured by depression, anxiety, cognitive difficulties, and neurobehavioral symptoms. Endorsements of general health problems, motor symptoms, and psychiatric history were also collected. Age of first exposure groups was compared using t tests, $\chi 2$ tests, and multivariable linear regressions, which included the following covariates: age, number of prior concussions, and total years of contact sport.

Results: Of 69 men aged 70.5 ± 8.0 years, approximately one-third of the sample (34.8%) reported AFE-CS before age 12 years. That group had more years of contact sports (10.8 ± 9.2 years) compared with those with AFE-CS ≥12 (5.6 ± 4.5 years; P = 0.02). No differences were found after univariate testing between AFE-CS groups on all outcomes (P-values >0.05). Multivariable models suggest that AFE-CS is not a predictor of depression or anxiety. Those in the AFE-CS <12 group had fewer cognitive difficulties (P = 0.03) and fewer neurobehavioral symptoms (P = 0.03).

Conclusions: Those with AFE-CS <12 to contact sports did not have worse long-term brain health outcomes compared with those with AFE-CS ≥12. Individuals with AFE-CS <12 had significantly lower British Columbia Cognitive Complaints Inventory and Neurobehavioral Symptom Inventory scores compared with those with AFE-CS ≥12.

Clinical relevance: The benefits of earlier AFE-CS may outweigh the risks of head strikes and result in comparable long-term brain health outcomes.

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Antidoping testing and testicle cancer diagnosis: An unexpected helper. Journal of Science and Medicine in Sport, Ahead of Print: 1-3, 2024.

Abstract - Human chorionic gonadotrophin (hCG) is prohibited in male athletes due to its potential performance-enhancing effects. Some athletes with elevated hCG levels during routine doping tests have been diagnosed with testicular cancer, highlighting the inadvertent role of anti-doping screening in cancer detection. This study explores the incidental detection of testicular cancer through routine anti-doping tests among elite athletes, aiming to raise awareness about the potential secondary benefits of these screenings. We have analyzed cases from the past decade to demonstrate how early detection has facilitated timely medical interventions, allowing athletes to heal and return to competitive sports. Through these instances, athletes become unintentional advocates, contributing to public health awareness about this topic.

Artificial intelligence applications in the football codes: A systematic review. Journal of Sports Sciences, Ahead of Print: 1-17, 2024.

Abstract - Artificial Intelligence (AI) is increasingly being adopted across many domains such as transport, healthcare, defense and sport, with football codes no exception. Though there is a range of potential benefits of AI, concern has also been expressed regarding potential risks. An important first step in ensuring that AI applications in football are usable, beneficial, safe and ethical is to understand the current range of applications, the AI models adopted and their proposed functions. This systematic review aimed to identify different applications of AI across football codes to synthesize current knowledge and determine whether potential risks are being considered. The systematic review included 190 peer-reviewed articles. Nine areas of application were found ranging from athlete evaluation and event detection to match outcome prediction and injury detection and prediction. In total, 27 different AI models were identified, with artificial neural networks the most frequently applied. Five AI assessment metrics were identified including specificity, recall, precision, accuracy and F1-score. Four potential risks were identified, concerning data security, usability, data biases and inappropriate athlete load management. It is concluded that, though a wide range of AI applications currently exist, further work is required to develop AI for football and identify and manage potential risks.

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Association of genetic profile with muscle mass gain and muscle injury prevention in professional football (soccer) players after creatine supplementation. Nutrients, 16: 2511, 2024.

Abstract - Background: In recent years, the study of creatine supplementation in professional athletes has been of great interest. However, the genetics involved in response to supplementation is unknown. The aim of this study was to analyze, for the first time, the relationship between muscle performancerelated genes and the risk of an increased body mass index (BMI) and muscle mass and a decrease in fat mass in professional football players after creatine supplementation. Methods: For this longitudinal study, one hundred- and sixty-one-men's professional football players were recruited. The polymorphisms ACE I/D, ACTN3 c.1729C>T, AMPD1 c.34C>T, CKM c.*800A>G, and MLCK (c.49C>T and c.37885C>A) were genotyped using Single-Nucleotide Primer Extension (SNPE). To assess the combined impact of these six polymorphisms, a total genotype score (TGS) was calculated. The creatine supplementation protocol consisted of 20 g/day of creatine monohydrate for 5 days (loading dose) and 3–5 g/day for 7 weeks (maintenance dose). Anthropometric characteristics (body mass index (BMI), fat, and muscle mass) were recorded before and after the creatine supplementation protocol. Characteristics of non-contact muscle injuries during the 2022/2023 season were classified according to a consensus statement for injury recording. The results showed that the allelic frequencies of ACE and AMPD1 differed between responders and non-responders in muscle mass increase (all $p < 0.05$). Players with a TGS exceeding 54.16 a.u. had an odds ratio (OR) of 2.985 (95%CI: 1.560–5.711; $p = 0.001$) for muscle mass increase. By contrast, those with a TGS below 54.16 a.u. had an OR of 9.385 (95%CI: 4.535–19.425; $p < 0.001$) for suffering non-contact muscle injuries during the season. Conclusions: The increase in BMI and muscle mass in response to creatine supplementation in professional football players was influenced by a TGS derived from the combination of favorable genotypes linked to muscle performance. The CC genotype and C allele of AMPD1 were particularly associated with a higher likelihood of muscle mass increase under creatine supplementation in this group of professional football players.

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Athlete monitoring perspectives of sports coaches and support staff: A scoping review. International Journal of Sports Science & Coaching, Ahead of Print: 1-20, 2024.

Objectives: To map and summarize the sports coaches' and support staff's perspectives on athlete monitoring to explore the breadth of literature, identify knowledge gaps and inform future research. Design: Scoping review based on the Joanna Briggs Institute (JBI) methodology.

Methods: SPORTDiscus, MEDLINE, APA PsycInfo, and Embase databases were searched in English until 6 September 2022. The inclusion criteria were (1) coach(es) and/or support staff were explicitly questioned about their knowledge, perceptions, understanding, opinions, and/or applied practice of athlete monitoring; (2) results could be directly attributed to coach(es) and/or support staff; (3) primary research projects that are available as full-text. Exclusion criteria were applied for grey literature. The data were extracted into a custom-made data charting spreadsheet.

Results: From the 4381 identified records, 42 met the eligibility criteria. Almost all the studies were conducted within the Anglosphere and at the national or international level. The main reasons for coaches and support staff to implement athlete monitoring were to reduce injury and illness, inform the training program, and improve or maintain performance. While training load monitoring is generally seen as valuable the coaches and support staff acknowledged that there was no perfect scientific approach to monitoring athletes and believed it should be part of the bigger picture, emphasizing communication.

Conclusions: There has been a recent surge in research demonstrating that athlete monitoring extends beyond quantitative information and encompasses non-quantified subjective information. This further substantiates that coaches and support staff will remain central to athlete monitoring, even amidst the anticipated technological progress.

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Between-microcycle variability of external soccer training loads through the evaluation of a contemporary periodization training model 'CUPs'. International Journal of Sports Science & Coaching, Ahead of Print: 1-11, 2024.

Objective: Variation in training load is consistently demonstrated within weekly microcycles in soccer, yet less is known of load variations between the same weekday sessions across different microcycles. Our study aim was to examine between-microcycle variability in key measures of external training load.

Methods: Thirty-seven professional soccer players participated in this observational study which took place across the clubs' initial 8-week in-season mesocycle of the 2022/23 season. During this mesocycle, each 1-week microcycle consisted of four distinct classifications of training session (Matchday (MD)-4, MD-3, MD-2, and MD-1, and one match (Saturday). External load data (total distance, high-intensity (>5.5 m.s) distance, high-intensity accelerations (>3 m/s^2), and percentage (%) of maximal speed attained) were collected across 564 training sessions (MD-4 =123, MD-3 =148, MD-2 =130, MD-1 =163). Data were analysed with mixed linear modelling.

Results: When compared to the first microcycle, substantial week-to-week variation was evident for each of the four training session classifications, ranging from 1244 m to 2248 m for total distance, 80 m to 197 m for high-intensity distance, 11 to 25 for high-intensity accelerations, and 10.2 percentage points to 15.4 percentage points for % maximal speed.

Conclusion: Our data show that despite training sessions having a consistency of planned training stimulus across an 8- week mesocycle, external load varied between microcycles. Nevertheless, within-player variability on the same day relative to matchday indicated a more consistent stimulus for key training variables relevant to specific training days.

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Blood, Sweat, and Tears: Implications for hydration testing in combat sports – Investigating body mass loss and biomarker changes. Scandinavian Journal of Medicine & Science in Sports, 34:e14701, 2024.

Abstract - Combat sports athletes often undergo rapid body mass loss (BML), which presents health risks. Hydration testing has been proposed as a possible solution to reduce or eliminate rapid BML. However, combat sports athletes may exhibit distinct physiological characteristics due to repeated exposure to BML. Thus, traditional and emerging hydration biomarkers should be investigated to determine their potential suitability for field use in this cohort. This study examined whether BML can explain changes in serum and urine osmolality (SosmΔ, UosmΔ), tear osmolarity (TosmΔ), hematocrit (HctΔ), and urine-specific gravity (USGΔ) after mild–moderate passive dehydration. Biomarker reliability was also assessed across two trials. Fifteen male and female combat sports athletes (age: 26.3±5.3 years, body mass: 67.7±9.9kg) underwent a sauna protocol twice (5–28days apart) aiming for 4% BML. The average BML in Trials 1 and 2 was 3.0±0.7%. Regression analysis revealed that BML explained HctΔ ($R2=0.22$, $p=0.009$) but not SosmΔ ($R2=0.11$, $p=0.079$) or other biomarkers. Intraclass correlation coefficients (ICCs) were significant for all biomarkers except TosmΔ (ICC=0.06, $p=0.37$) and post-Tosm (ICC=0.04, $p=0.42$); post-Hct performed best (ICC=0.82, $p<0.001$). Contingency tables with post-Sosm (295mOsm/kg) and post-USG (1.020) cutoffs revealed an 80% true negative rate (TNR) and a 62% true positive rate (TPR). Increasing the Sosm cutoff to 301mOsm/kg decreased the TNR to 52% but increased the TPR to 83%. Although blood parameters were most sensitive to BML, they could only explain 11%–22% of biomarker variation. The typical USG cutoff misclassified 42% of athletes postdehydration, and reliability was generally poor-moderate. Alternative strategies should be pursued to manage rapid BML in combat sports.

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Bye-Bye, Bye Advantage: Estimating the competitive impact of rest differential in the National Football League. arXiv:2408.10867, 2024.

Abstract - The National Football League (NFL) sets its regular season schedule to optimize viewership and minimize competitive inequities. One inequity assumed to impact team performance is rest differential, defined as the relative number of days between games. Using Bayesian state space models on both game outcomes and betting market data, we estimate the competitive effect of rest differential in American football. We find that the most commonly referred to inequities -- both the bye week rest advantage and the mini-bye week rest advantage -- currently show no significant evidence of providing the rested team a competitive edge. Further, we trace a decline in the advantage of a bye week to a 2011 change to the NFL's Collective Bargaining Agreement, which represents a natural experiment to test the relevance of rest and preparation in football. Prior to the agreement, NFL teams off a bye week received a significant advantage (+2.2 points per game), but since 2011, that benefit has been mitigated.

Can caffeine change the game? Effects of acute caffeine intake on specific performance in intermittent sports during competition: A systematic review and meta-analysis. International Journal of Sports Physiology and Performance, Ahead of Print: 1-17, 2024.

Background: The benefits of oral caffeine intake to enhance several aspects of physical performance, such as aerobic endurance, strength, power, and muscle endurance performance, are well supported. However, how the physical performance benefits of caffeine supplementation are translated into better specific actions in intermittent sports during real or simulated competition has been the topic of fewer investigations, and their results need to be appropriately reviewed and meta-analyzed.

Objective: The aim of the study was to investigate the effects of acute caffeine intake on specific actions in intermittent sports involving decision making and high-intensity efforts (eg, team, racket, and combat sports) during real or simulated competitions.

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Methods: All studies included had blinded and crossover experimental designs, and we conducted a risk-of-bias analysis. In total, we included 24 studies. A meta-analysis was performed using the random-effects model to calculate the standardized mean difference (SMD) estimated by Hedges g and 95% CIs.

Results: Caffeine ingestion increased high-intensity sport-specific actions during competition, such as the number of sprints (SMD: 0.48; 95% CI, 0.23–0.74), body impacts (SMD: 0.28; 95% CI, 0.08– 0.49), accelerations (SMD: 0.35; 95% CI, 0.06–0.63), decelerations (SMD: 0.63; 95% CI, 0.12–1.14), and high-intensity offensive efforts (SMD: 0.36; 95% CI, 0.11–0.61). Additionally, caffeine ingestion induced a higher positive or success rate of actions during real or simulated competition (SMD: 0.44; 95% CI, 0.19–0.69).

Conclusion: The current meta-analysis provides evidence of caffeine supplementation in increasing high-intensity efforts and the success rate of sport-specific actions during real or simulated competition.

Cannabinoid Therapy in Athletics: A review of current cannabis research to evaluate potential real-world cannabinoid applications in sport. Sports Medicine, Ahead of Print: 1-27, 2024.

Abstract - The increasing legalization of Cannabis sativa plant products has sparked growing interest in their therapeutic applications. Prohibition laws established in 1937 hindered formal research on cannabis, a plant with cultural and medicinal roots dating back to 2700 BC in Chinese history. Despite regulatory hurdles, published research on cannabis has emerged; yet elite athletes remain an underrepresented population in these studies. Athletes, known for exploring diverse substances to optimize performance, are drawn to the potential benefits of cannabinoid therapy, with anecdotal reports suggesting positive effects on issues ranging from anxiety to brain injuries. This review aims to evaluate empirical published cannabis research with a specific focus on its potential applications in athletics. The changing legal landscape, especially the removal of cannabis from drug testing programs in leagues such as the National Basketball Association (NBA), and endorsements by Major League Baseball (MLB) for cannabinoid products and the National Football League (NFL) for cannabis research, reflects a shift in the acceptability of such substances in sports. However, stigma, confusion, and a lack of education persist, hindering a cohesive understanding among sports organizations, including business

professionals, policymakers, coaches, and medical/training staff, in addition to athletes themselves. Adding to the confusion is the lack of consistency with cannabinoid regulations from sport to sport, within or out of competition, and with cannabis bioactive compounds. The need for this review is underscored by the evolving attitudes toward cannabinoids in professional sports and the potential therapeutic benefits or harms they may offer. By synthesizing current cannabis research, this review aims to provide a comprehensive understanding of the applications and implications of cannabinoid use in the realm of athletics.

Classification of myo-connective tissue injuries for severity grading and return to play prediction: A scoping review. Journal of Science and Medicine in Sport, Ahead of Print: 1-40, 2024.

Objectives: To conduct a systematic literature search to identify currently used classifications of acute non -contact muscle injuries in sporting adults.

Design: Scoping review.

Methods: A systematic literature search from January 1, 2010, to April 19, 2022 of Medline and SPORTDiscus yielded 13 426 articles that were screened for eligibility. Findings from included studies were qualitatively synthesized. Classifications and their grading, as well as outcomes and definitions were extracted.

Results: Twenty -four classifications were identified from the 37 included studies, most of which had low evidence study designs. Majority (57%) of classifications were published after 2009 and were mostly developed for hamstring or other lower limb injuries. The six most cited classifications accounted for 70% of the reports (BAMIC, modified Peetrons, Munich, Cohen, Chan and MLG -R). Outcome reporting was sparse, making it difficult to draw conclusions. Still, significant relationships between grading and time to return to play were reported for the BAMIC, modified Peetrons, Munich and Cohen classifications. Other classifications either had a very low number of reported associations, reported no associations, reported inconclusive associations, or did not report an assessment of the association. Other outcomes were poorly investigated.

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Conclusion: There is no agreed -upon use of muscle classification, and no consensus on definitions and terminology. As a result, reported outcomes and their relationship to severity grading are inconsistent across studies. There is a need to improve the generalizability and applicability of existing classifications and to refine their prognostic value. High -level evidence studies are needed to resolve these inconsistencies.

Comparing the maximal horizontal deceleration demands between a novel acceleration to deceleration assessment and the 505 change of direction test. Journal of Human Sport and Exercise, 19(3): 803-814, 2024.

Abstract - The importance of quantifying maximal horizontal deceleration performance in athlete populations has received a considerable increase in interest over recent years. However, research is still scarce investigating movement characteristics of maximal horizontal decelerations outside of measures derived from instantaneous horizontal velocity of the centre of mass, using technologies such as radar or laser-based devices. Therefore, this study aimed to explore the biomechanical differences for measures of deceleration ability between a novel deceleration task, and the 505 change of direction test, using an inertial measurement unit-based technology. Primary findings suggested differences across several biomechanical characteristics quantified during the deceleration phase, with moderate to large between-test effect sizes. Specifically, subjects were found to exhibit significantly greater reductions in velocity and horizontal braking forces in the 505. Further, subjects showed significantly shorter stopping times and distances in the acceleration-deceleration assessment, however, these displayed insufficient levels of reliability across both assessments, which should be interpreted as a limitation. Therefore, it may be speculated that based on our data, the 505 test, which possesses a predetermined stopping/turning point, presents a greater or different biomechanical challenge to individuals, which must be met with the appropriate neuromuscular and skill-related qualities to efficiently reduce whole-body momentum. These findings may be relevant to practitioners interested in choosing the right assessment to quantify athletes' maximal horizontal deceleration performance, which can have implications for both health and performance.

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Correlation of player and imaging characteristics with severity and missed time in National Football League professional athletes with hamstring strain injury – A retrospective review. American Journal of Sports Medicine, Ahead of Print: 1-9, 2024.

Background: Hamstring strain injuries (HSIs) are prevalent in US National Football League (NFL) players, but there is a paucity of information regarding imaging characteristics, injury severity, and player factors associated with time missed and risk of recurrent injury.

Purpose: To describe player, football activity, clinical, and imaging characteristics of NFL players with HSIs, as well as determine player characteristics, clinical examination results, and magnetic resonance imaging (MRI) findings associated with injury occurrence, severity, and missed time.

Study design: Cross-sectional study; Level of evidence, 3.

Methods: A retrospective cohort of NFL players with acute HSI (n = 180) during the 2018-2019 season was identified. Injury data were collected prospectively through a league-wide electronic health record system. Three musculoskeletal radiologists graded MRI muscle injury parameters using the British Athletics Muscle Injury Classification (BAMIC) system. Player, football, clinical, and imaging characteristics were correlated with HSI incidence and severity and with missed time from sport.

Results: Of the 1098 HSIs identified during the 2018-2019 season, 416 (37.9%) were randomly sampled, and 180 (43.3%) had diagnostic imaging available. Game activity, preseason period, and wide receiver and defensive secondary positions disproportionately contributed to HSI. The biceps femoris was the most commonly injured muscle (n = 132, 73.3%), followed by the semimembranosus (n = 24, 13.3%) and semitendinosus (n = 17, 9.4%) muscles. The most common injury site was the distal third of the biceps femoris and semitendinosus muscles (n = 60, 45.5% and n = 10, 58.8%, respectively) and central part of the semimembranosus muscle (n = 17, 70.8%). Nearly half of the injuries (n = 83, 46.1%) were BAMIC grade 2; 25.6% (n = 46), grade 3; and 17.8% (n = 32), grade 4. MRI showed sciatic nerve abnormality in 30.6% (n = 55) of all HSIs and 81.3% (n = 26) of complete tendon injuries. BAMIC grade correlated with both median days and games missed. Combined biceps femoris and semitendinosus injuries resulted in the highest median days missed (27 days).

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Conclusion: Among NFL players with acute HSIs, the most common injury was a moderate-severity injury of the distal biceps femoris. BAMIC grade was associated with missed time.

Does draft currency promote competitive balance? An empirical investigation of the National Football League 2002-2021. Journal of Sports Economics, Ahead of Print: 1-23, 2024.

Abstract - In the National Football League (NFL) annual draft, teams take turns selecting entering players. The draft is a market mechanism designed to promote competitive balance as the NFL assigns draft positions to teams in reverse order relative to last season's performance. Teams frequently trade draft picks for other picks and/or players. We use several market valuations of draft picks to define original draft currency as the total value of draft picks available before any trades and final draft currency as the total value of picks used after all trades. For the 2002–2021 period, we find that original draft currency does not affect the probability of reaching the playoffs, but final draft currency does. Usage of outdated market valuations of draft picks by most teams can help explain how the best team has used draft-pick trades to remain among the strongest teams over two decades thereby perpetuating competitive imbalance.

Effect of data-processing methods on acceleration summary metrics of GNSS devices in elite Australian football. Sensors, 24: 4383, 2024.

Abstract - This study aimed to measure the differences in commonly used summary acceleration metrics during elite Australian football games under three different data processing protocols (raw, custom-processed, manufacturer-processed). Estimates of distance, speed and acceleration were collected with a 10-Hz GNSS tracking technology device from fourteen matches of 38 elite Australian football players from one team. Raw and manufacturer-processed data were exported from respective proprietary software and two common summary acceleration metrics (number of efforts and distance within medium/high-intensity zone) were calculated for the three processing methods. To estimate the effect of the three different data processing methods on the summary metrics, linear mixed models were

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used. The main findings demonstrated that there were substantial differences between the three processing methods; the manufacturer-processed acceleration data had the lowest reported distance (up to 184 times lower) and efforts (up to 89 times lower), followed by the custom-processed distance (up to 3.3 times lower) and efforts (up to 4.3 times lower), where raw data had the highest reported distance and efforts. The results indicated that different processing methods changed the metric output and in turn alters the quantification of the demands of a sport (volume, intensity and frequency of the metrics). Coaches, practitioners and researchers need to understand that various processing methods alter the summary metrics of acceleration data. By being informed about how these metrics are affected by processing methods, they can better interpret the data available and effectively tailor their training programs to match the demands of competition.

Effect of soccer boot outsole configuration on translational traction across both natural and artificial playing surfaces. The Orthopaedic Journal of Sports Medicine, 12(8): 1-10, 2024.

Background: Soccer boots are produced with different stud patterns and configurations to provide players with extra traction on specific surface types to minimize slipping and improve player performance. Excessive traction, however, can lead to foot fixation injuries, particularly anterior cruciate ligament tears.

Purpose/hypothesis: The purpose of this study was to explore the translational traction properties of 5 different outsole configurations moving in 4 different directions across both natural grass and artificial grass (AG) playing surfaces. It was hypothesized that longer studs or studs with an asymmetric shape would yield a higher traction coefficient compared with the recommended stud configuration for the given playing surface.

Study design: Descriptive laboratory study.

Methods: A custom-built testing apparatus recorded the translational traction of 5 different soccer boots moving in an anterior, posterior, medial, or lateral direction on both natural grass and AG playing surfaces. A 3-way analysis of variance was performed to determine the effect of outsole configuration

on the traction, and a post hoc Tukey analysis was performed to compare different outsole configurations with a control.

Results: For the natural grass playing surface, the longer and asymmetric studs yielded a significantly higher ($P < .05$) traction coefficient on 75% of loading scenarios, while on AG, they yielded a significantly higher traction on 50% of loading scenarios.

Conclusion: Some soccer boots yielded higher traction values compared with the recommended configuration.

Clinical relevance: The results highlight the importance of boot selection on different playing surfaces. Higher traction values could increase the injury risk for players due to excessive traction and foot fixation.

Effects of acute sleep deprivation on sporting performance in athletes: A comprehensive systematic review and meta-analysis. Nature and Science of Sleep, 16: 935-948, 2024.

Objective: Using meta-analysis to comprehensively and quantitatively evaluate the impact of acute sleep deprivation on different sports performance of athletes, this study aims to provide scientific guidance for coaches in optimizing and adjusting training and competition arrangements.

Methods: Establishing literature inclusion and exclusion criteria, we conducted searches in both Chinese and English databases. Using stata 14.0, we analyzed 75 indicators from 27 included literature, focusing on three aspects: the impact of acute sleep deprivation on overall athletic performance, the impact on sporting performance across various athletic abilities, and the disparities in athletic performance between morning and afternoon following acute sleep deprivation.

Results: The effect size of acute sleep deprivation on overall athletic performance was -0.56 ($P<0.05$). Sub-analyses revealed effect sizes of -0.23 ($P<0.05$) for whole night sleep deprivation, -1.17 ($P<0.05$) for partial sleep deprivation at the end of the night, and -0.25 ($P>0.05$) for partial sleep deprivation in the beginning of the night. The effect sizes of acute sleep deprivation on high intensity intermittent exercise,

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skill control, speed, aerobic endurance, and explosive power indicators were -1.57, -1.06, -0.67, -0.54, and -0.39 respectively (P<0.05). The effect sizes of acute sleep deprivation on the overall athletic performance in the morning and afternoon were -0.30, and -1.11, respectively (P<0.05).

Conclusion: Acute sleep deprivation significantly impairs the overall athletic performance of athletes, with a more pronounced negative impact observed with partial sleep deprivation at the end of the night. Various types of exercise performance are adversely affected by acute sleep deprivation, with magnitude of impact ranking high intensity intermittent, skill control, speed, aerobic endurance, and explosive power. Following acute sleep deprivation, athletes' overall sporting performance in the afternoon is inferior to that in the morning.

Efficacy of the FIFA cooling break heat policy during an intermittent treadmill football simulation in hot conditions in trained males. British Journal of Sports Medicine, Ahead of Print: 1-8, 2024.

Objective: To evaluate the efficacy of the Fédération Internationale de Football Association (FIFA) cooling break policy against alternative cooling configurations in attenuating thermal strain during simulated football in the heat.

Methods: 12 males (age: 27±6 years, V̇O2peak: 61±7 mL/kg/min) completed five 90 min intermittent treadmill football match simulations in 40°C and 41% relative humidity (32°C wet-bulb globe temperature) with different cooling configurations: regular match without cooling breaks (REG), 3 min breaks without cooling (BRKno-cool), 3 min breaks with cooling (BRKcool: current FIFA policy; chilled fluid ingestion and ice towel across neck and shoulders), 5 min extended half-time without cooling breaks (ExtHTonly) and 3 min cooling breaks with 5 min ExtHT (ExtHTcool). Rectal temperature (Tre), heart rate, whole-body sweat rate (WBSR) and rating of perceived exertion (RPE) were recorded. Data are presented as mean (95% CIs).

Results: Final Tre was lower in BRKno-cool (0.20°C (0.01, 0.39), p=0.038), BRKcool (0.39°C (0.21, 0.57), p<0.001) and ExtHTcool (0.40°C (0.22, 0.58), p<0.001) than REG (39.1°C (38.8, 39.3)). Mean Tre was lower in ExtHTcool (38.2°C (38.0, 38.4)) than BRKcool (38.3°C (38.1, 38.5), p=0.018), BRKno-cool and

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ExtHTonly (38.4°C (38.2, 38.6), p<0.001) and REG (38.5°C (38.3, 38.7), p<0.001). Mean heart rate was lower during BRKcool (6 beats/min (4, 7), p<0.001) and ExtHTcool (7 beats/min (6, 8), p<0.001) compared with REG. WBSR was comparable across trials (p≥0.07) and RPE was attenuated during BRKcool (0.4 (0.1, 0.7), p=0.004) and ExtHTcool (0.5 (0.2, 0.7), p=0.002), compared with REG.

Conclusion: BRKcool and ExtHTcool attenuated thermal, cardiovascular and perceptual strain during a simulated football match in the heat. Additional strategies may be required in field settings or under harsher conditions.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Amateur and elite football players often play in hot and/or humid environments that lead to reductions in physical performance and increase the risk of exertional heat illness.

⇒ To reduce the adverse effects of playing football under environmental heat stress, the Fédération Internationale de Football Association (FIFA) Medical Committee implemented brief in-play cooling breaks when the outdoor wet-bulb globe temperature (WBGT) reaches 32°C.

WHAT THIS STUDY ADDS

⇒ This study evaluated the FIFA cooling break policy by implementing different cooling configurations during simulated matches at 32°C WBGT conditions, to determine the most effective approach for attenuating thermal strain in trained males. The current FIFA cooling break policy leads to a reduction in thermal, cardiovascular and perceptual strain, with the combination of the 3min cooling breaks and a 5min extension to half-time further reducing thermal and cardiovascular strain.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Findings from this study show that despite the cooling breaks and extended half-time attenuating physiological and perceptual strain during a simulated football match, future research may seek to evaluate different cooling break approaches in real-world settings.

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Enhancing athletic performance with complex contrast training: A Delphi study of elite strength and conditioning coaches. International Journal of Sports Science & Coaching, Ahead of Print: 1-13, 2024.

Abstract - This Delphi study identifies how elite strength and conditioning coaches prescribe complex contrast training to team sport athletes. Thirty-eight elite strength and conditioning coaches with experience working in high performance settings participated in a three-round survey. Round one was used to collect open-ended responses regarding the programming of complex contrast training. Responses were formulated into statements for rounds two and three, where participants rated their level of agreement using a 9-point Likert scale (agreement threshold ≥80%). In round one, 139 statements were formulated. Twenty-seven statements reached consensus in round two, and four more in round three. Based upon consensus, complex contrast training is best used during the season and dedicated power blocks. It was recommended that strength and power exercises within complex contrast training pairings be performed for 3–4 sets of 3– 4 repetitions. Loading of 85–90% of one repetition maximum (1RM) was recommended for strength exercises, with ~two repetitions in reserve, and BW–30% 1RM for power exercises, with three minutes of rest between contrast sets. Upper body pressing, upper body throwing, lower body squatting and hinging, and lower body jumping and bounding exercises were recommended. Coaches reported using complex contrast training to enhance power-focused training, and because it is more time efficient. Complex contrast training may be a time efficient method of improving athletic performance when training time is limited. Results from this study can guide the development of complex contrast training programs for team-sport athletes, and provides valuable insight into how elite coaches enhance performance.

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Examining the prevalence of anterior cruciate ligament injuries on artificial turf surfaces compared to natural grass surfaces in athletes: A scoping review. Cureus, 16(7): e63779, 2024.

Abstract - The anterior cruciate ligament (ACL) is commonly injured in sports such as American football and soccer. It is currently unknown if ACL injuries are more prevalent on natural grass or artificial turf fields. The purpose of this scoping review is to analyze research studies evaluating the effect of the playing surface on the prevalence of ACL injuries. We hypothesize that athletes face a greater risk of suffering ACL injuries while playing on artificial turf compared to natural grass. Our team conducted a comprehensive literature review by screening three databases (PubMed, Embase, and Cochrane) that comprised a wide range of peerreviewed articles on ACL injuries suffered on natural grass and artificial turf surfaces. Inclusion criteria consisted of epidemiological and cohort studies published after 1990 that were written in English and focused on athletes ranging in skill level from youth to professional. Exclusion criteria consisted of biomechanical studies, review articles, and papers that focused on injuries of structures other than the ACL. Bias was assessed with the MINORS criteria. Results were presented by injury rates, calculated ratios, and confidence intervals. The final analysis included nine papers published in peer-reviewed journals. Three of nine papers found that ACL injuries are more likely to occur on artificial turf than on natural grass. Three papers found that there is no difference in the prevalence of ACL injuries between surfaces and one paper stated that ACL injuries are more likely on natural grass than artificial turf. Two papers did not report confidence intervals for ratios comparing injury rates between playing surfaces. There is no consensus in the current literature regarding the prevalence of ACL injuries on artificial turf versus natural grass surfaces. The primary limitation of this study was that the papers used a variety of methods to compare rates of ACL injuries on artificial and natural surfaces, making comparisons between the nine papers difficult.

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External training loads and soft-tissue injury occurrence during congested versus non-congested periods in football (soccer). International Journal of Sports Physiology and Performance, Ahead of Print: 1-8, 2024.

Objective: To analyze the influence of congested and noncongested fixture periods during 2 seasons in a professional male football team on soft-tissue injury incidence and external load.

Methods: Thirty-three professional football players from a Portuguese Liga I team participated in this study. Weekly external load and soft-tissue injury rate and burden of 2 consecutive seasons (2021–22 and 2022–23) were analyzed.

Results: Total soft-tissue injury rate and burden for the 2 seasons were 3.9 and 3.2 injuries per 1000 hours and 71.8 and 60.5 days per 1000 hours for congested and noncongested periods, respectively. No significant differences were observed between congested and noncongested periods. Total high-speed running, sprint distance, distance above 80% and 90% of maximal velocity, and meters accelerating and decelerating above 2 m/s2 were significantly higher for noncongested weeks. Match accelerations and decelerations above 3 m/s2 were higher during congested periods and training during noncongested periods. No differences between the 2 periods were observed for the total number of accelerations and decelerations above 3 m/s2 . Overall, physical outputs per week were higher for training during noncongested weeks, whereas matches during congested periods registered higher external load.

Conclusions: No effect of a congested schedule was observed on soft-tissue injury rates and injury burden. Higher match exposure during congested periods increased external load performed per week, and during noncongested periods, training load was superior to congested weeks.

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Hamstring muscle stiffness in athletes with and without anterior cruciate ligament reconstruction history: A retrospective study. Journal of Clinical, 13: 4370, 2024.

Introduction: Sports requiring sprinting, jumping, and kicking tasks frequently lead to hamstring strain injuries (HSI). One of the structural risk factors of HSI is the increased passive stiffness of the hamstrings. Anterior cruciate ligament (ACL) injury history is associated with a 70% increase in the incidence of HSI, according to a recent meta-analysis. The same report recommended that future research should concentrate on the relationships between the HSI risk factors. Hence, the present study aimed to retrospectively compare changes in the passive stiffness of the hamstrings in athletes with and without ACL reconstruction history.

Methods: Using ultrasound-based shearwave elastography, the mid-belly passive muscle stiffness values of the biceps femoris long head, semimembranosus, and semitendinosus muscles were assessed and compared amongst athletes with and without a history of ACL reconstruction.

Results: There were no significant differences in the biceps femoris long head (injured leg (IL): 26.19 ± 5.28 kPa, uninjured contralateral (UL): 26.16 ± 7.41 kPa, control legs (CL): 27.64 ± 5.58 kPa; IL vs. UL: $p = 1$; IL vs. CL: $p = 1$; UL vs. CL: $p = 1$), semimembranosus (IL: 24.35 ± 5.58 kPa, UL: 24.65 ± 8.35 kPa, CL: 22.83 ± 5.67 kPa; IL vs. UL: $p = 1$; IL vs. CL: $p = 1$; UL vs. CL, $p = 1$), or semitendinosus (IL: 22.45 ± 7 kPa, UL: 25.52 ± 7 kPa, CL: 22.54 ± 4.4 kPa; IL vs. UL: $p = 0.487$; IL vs. CL: $p = 1$; UL vs. CL, $p = 0.291$) muscle stiffness values between groups.

Conclusions: The passive mid-muscle belly stiffness values of the biceps femoris long head, semitendinosus, and semimembranosus muscles did not significantly differ between previously injured and uninjured athletes; therefore, further assessment for other muscle regions of hamstrings may be necessary. To collect more comprehensive data related to the structural changes that may occur following ACL reconstructions in athletes, a future study should examine the passive stiffness of wider muscle regions from origin to insertion.

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Hamstrings on focus: Are 72 hours sufficient for recovery after a football (soccer) match? A multidisciplinary approach based on hamstring injury risk factors and histology. Journal of Sports Sciences, Ahead of Print: 1-18, 2024.

Abstract - This study aimed to assess acute and residual changes in sprint-related hamstring injury (HSI) risk factors after a football (soccer) match, focusing on recovery within the commonly observed 72-h timeframe between elite football matches. We used a multifactorial approach within a football context, incorporating optical and ultrastructural microscopic analysis of BFlh (biceps femoris long head) muscle fibres, along with an examination of BFlh fibre composition. Changes in sprint performance-related factors and HSI modifiable risk factors were examined until 3 days after the match (MD +3) in 20 football players. BFlh biopsy specimens were obtained before and at MD +3 in 10 players. The findings indicated that at MD +3, sprint-related performance and HSI risk factors had not fully recovered, with notable increases in localized BFlh fibre disruptions. Interestingly, match load (both external and internal) did not correlate with changes in sprint performance or HSI risk factors nor with BFlh fibre disruption. Furthermore, our study revealed a balanced distribution of ATPase-based fibre types in BFlh, with type-II fibres associated with sprint performance. Overall, the results suggest that a 72-h recovery period may not be adequate for hamstring muscles in terms of both HSI risk factors and BFlh fibre structure following a football match.

How humans run faster: The neuromechanical contributions of functional muscle groups to running at different speeds. Scandinavian Journal of Medicine & Science in Sports, 34:e14690, 2024.

Abstract - How the neuromechanics of the lower limb functional muscle groups change with running speed remains to be fully elucidated, with implications for our understanding of human locomotion, conditioning, and injury prevention. This study compared the neuromechanics (ground reaction and joint kinetics, kinematics and muscle activity) of middle-distance athletes running on an instrumented treadmill at six wide-ranging speeds (2.78–8.33 $m \cdot s^{-1}$). Ground reaction forces and kinematics were analyzed using inverse dynamics to calculate flexor and extensor joint torques, and positive and negative work done by these torques. Contributions of each functional muscle group to the total

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positive and negative work done by the limb during stance, swing, and the whole stride were quantified. During stance, the ankle plantar flexors were the major energy generator and absorber (>60%) at all speeds, but their contribution to whole stride energy generation and absorption declined with speed. Positive work by the hip extensors rose superlinearly with speed during stance (3-fold) and especially during swing (12-fold), becoming the biggest energy generator across the whole stride at >5 m·s−1. Knee flexor and extensor negative work also rose superlinearly with speed during swing, with the knee flexors becoming the greatest energy absorber over the whole stride at >7.22 $m \cdot s^{-1}$. Across speeds, plantar flexor peak moment and positive work accounted for 97% and 96% of the variance in step length, and swing hip extension peak moment and positive work accounted for 98% and 99% of the variance in step frequency. There were pronounced speed, phase (stance/swing), and work (positive/negative) dependent contributions of the different functional muscle groups during running, with extensive implications for conditioning and injury prevention.

Lower extremity return to sport testing: A systematic review. The Knee, 50: 115-146, 2024.

Background: Lower extremity injuries account for an enormous portion of sports medicine cases in the United States each year. Unfortunately, there are no uniform criteria for athletes to complete prior to returning to sport (RTS) following a lower extremity injury. Therefore, the purpose of this systematic review is to review current literature for joint-specific and global lower extremity testing to determine the most valid functional test that can be utilized to reduce the risk of re-injury as athletes RTS.

Methods: A systematic search of PubMed, PubMed Central, Cochrane Library, OVID, and Embase databases was conducted for studies prior to May 2024 following PRISMA guidelines. ROBINS-I Tool was utilized for the risk of bias assessment.

Results: Of 19,189 studies, 114 (0.6%) studies published prior to May of 2024 met inclusion criteria and were analyzed. Eighty five percent of articles discussed RTS for individuals with knee pathology. Furthermore, 82% specifically analyzed RTS following ACL reconstruction. The most common RTS test was isokinetic dynamometry testing which is seen in 73% of studies. Only 6.2% of studies analyzed RTS for individuals with hip pathology and only two studies analyzed RTS for patients with ankle injuries.

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Conclusion: Even with the enormous amount of literature that exists regarding ACL injuries and testing there is no standardized criterion for RTS clearance. The suggested test batteries from this review can serve as a framework for future research and validation for joint-specific RTS functional testing.

Meniscus tears in elite athletes: Treatment considerations, clinical outcomes, and return to play. Current Reviews in Musculoskeletal Medicine, 17: 313-320, 2024.

Purpose of review: Management of meniscal injuries in the elite athlete is a difficult problem secondary to the high demands of athletic competition, the need for a timely return to sport, and the desire to maximize performance over time. The purpose of this review is to provide an up-to-date summary on the current literature and trends regarding the management of meniscus injuries with a special consideration for elite athletes.

Recent findings: Historically, partial meniscectomy has been the primary treatment option for meniscus injuries. However, in recent years there has been an increased emphasis on meniscus preservation due to the increased risk of cartilage degeneration over time. Moreover, while partial meniscectomy still provides a quicker return to sport (RTS), recent literature has demonstrated similar rates of RTS and return to pre-injury levels between partial meniscectomy and meniscus repair. In the setting of symptomatic meniscal deficiency, meniscus allograft transplantation has become an increasingly utilized salvage procedure with promising yet variable outcomes on the ability to withstand elite competition. Currently, there is no uniform approach to treating meniscal injuries in elite athletes. Therefore, an individualized approach is required with consideration of the meniscus tear type, location, concomitant injuries, athlete expectations, rehabilitation timeline, and desire to prevent or delay knee osteoarthritis. In athletes with anatomically repairable tears, meniscus repair should be performed given the ability to restore native anatomy, provide high rates of RTS, and mitigate long-term chondral damage. However, partial meniscectomy can be indicated for unrepairable tears.

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Mental Fatigue in Sport – From impaired performance to increased injury risk. International Journal of Sports Physiology and Performance, Ahead of Print: 1-9, 2024.

Abstract - The literature describing the effects of mental fatigue (MF) has grown tremendously. This is accompanied by identification of a host of performance-determining parameters affected by MF. MF results from prolonged cognitive effort and predominantly affects physical, technical, tactical, and perceptual–cognitive dimensions of sport, while physiological parameters (eg, heart rate, lactate) and physical aspects of maximal and supramaximal efforts are predominantly unaffected. The aim of this paper was to provide an overview of the parameters described in the literature as influenced by MF. By identifying the different parameters, we not only see how they affect the performance of athletes but also raise concerns about the potentially increased injury risk due to MF. Preliminary evidence suggests that subsequent disturbances in balance, motor skills, and decision-making processes could potentially increase the vulnerability to injury. An abundance of lab-based studies looked into the effects of MF on performance; however, many questions remain about the mechanisms of origin and neurophysiological causes of MF, and only small steps have been taken to translate this knowledge into practice. Thus, there is a need for more research into the underlying mechanisms of MF and the role of the brain, as well as more applied research with a high ecological validity that also takes into account the potential increased risk of injury due to MF.

Oral fluoroquinolones and the risk of Achilles tendon rupture. Journal of Sport and Health Science, Ahead of Print: 1-3, 2024. (No abstract available)

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Predicting sprint performance from the vertical and horizontal jumps in National Football League combine athletes. Journal of Strength and Conditioning Research, 38(8): 1433-1439, 2024.

Abstract - Identifying fast athletes is an important part of the National Football League (NFL) Combine. However, not all athletes partake in the 36.58-m sprint, and relying on this single test may miss potentially fast athletes. Therefore, the purpose of this study was to determine whether sprinting times can be predicted using simple anthropometric and jumping measures. Data from the NFL Combine between the years 1999-2020 inclusive were used (n = 4,149). Subjects had a mean (±SD) height = 1.87 ± 0.07 m and body mass = 111.96 ± 20.78 kg. The cross-validation technique was used, partitioning the data into a training set (n = 2,071) to develop regression models to predict time over the 9.14-, 9.14- to 18.29-, 18.29- to 36.58-m, and 36.58-m segments using vertical jump, broad jump, height, and mass as the independent variables. The models were then evaluated against a test set (n = 2,070) for agreement. Statistically significant ($p < 0.01$) models were determined for 9.14-m time (adjusted R2 = 0.76, SEE = 0.05 seconds), 9.14- to 18.29-m time (adjusted R2 = 0.74, SEE = 0.04 seconds), 18.29- to 36.59-m time (adjusted R2 = 0.79, SEE = 0.07 seconds), and 36.58-m time (adjusted R2 = 0.84, SEE = 0.12 seconds). When evaluated against the test set, the models showed biases of -0.05, -0.04, -0.02, and -0.02 seconds and root-mean-square error of 0.07, 0.05, 0.07, and 0.12 seconds for the 9.14-, 9.14- to 18.29-, 18.29- to 36.58-m, and 36.58-m segments, respectively. However, 5-6% of the predictions lay outside of the limits of agreement. This study provides 4 formulae that can be used to predict sprint performance when the 36.58-m sprint test is not performed, and practitioners can use these equations to determine training areas of opportunity when working with athletes preparing for the NFL Combine.

Practical Applications - This study provides 4 formulae that can be used to predict sprint performance using height, body mass, VJ, and BJ. This will allow practitioners to predict sprint time with incomplete sets of data and to identify athletes who may have possibly underperformed or overperformed in the 36.58-m sprint. Therefore, not every slow or fast athlete is equivalent, and those with superior jumping skills may still be draft candidates. Furthermore, practitioners can use these equations when performing a needs analysis to identify an athlete's strengths and weaknesses to improve sprinting performance.

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Preseason and in-season high-speed running demands of 2 professional Australian Rules Football teams. Sports Health, Ahead of Print: 1-12, 2024.

Background: Australian Rules Football athletes complete long preseasons, yet injuries occur frequently at early stages of the competitive season. Little is known about the high-speed running (HSR) prescription during a preseason or whether players are adequately prepared for competition. This study described absolute and relative preseason and in-season HSR demands of 2 professional Australian football teams.

Hypothesis: HSR and sprinting volumes are significantly lower in elite Australian Rules football athletes during in-season compared with preseason.

Study design: Cohort study.

Level of evidence: Level 3.

Methods: During the 2019 season, HSR volume was collected for 2 professional Australian football teams (n = 55). Individual maximum speeds (Vmax) were captured to calculate relative running speed thresholds, as reported in 5% increments from 70%Vmax to 100%Vmax.

Results: Weekly volume of running above 70%Vmax (P = 0.01; r = 0.56) and 80%Vmax (P = 0.01; r = 0.58) was significantly greater in the preseason than the in-season. The weekly volume completed above 90%Vmax was not significantly greater in the preseason than the in-season (P = 0.10; r = 0.22). Individual variation in the distance completed at specific percentages of Vmax expressed as a coefficient of variation was reported as 51% at 71% to 80%Vmax, 39% at 81% to 90%Vmax, and 41% at 91% to 100%Vmax.

Conclusion: The volume of HSR completed by athletes is far greater in the initial 4 weeks of the preseason than in any other point in preseason or competitive phases. At the individual level, there is substantial variation in the distance covered. This supports the concept of a heavily individualized approach to high-speed prescription and monitoring.

Clinical relevance: Practitioners should carefully consider individual variation regarding sprinting volume during both preseason and in-season when prescribing and monitoring training to improve on-field performance and reduce the risk of injury.

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Secondary injury prevention reduces hamstring strain and time-loss groin injury burdens in male professional football. Physical Therapy in Sport, Ahead of Print: 1-28, 2024.

Objectives: This study aimed to assess the effectiveness of a systems-based secondary injury prevention intervention aimed at early detection and management of hamstring strain injury (HSI) and time-loss groin injury in an Australian male professional football club.

Design: Prospective cohort study.

Setting: Australian male professional football club.

Participants: Data were collected from male professional football players (n=73) from a single football club.

Main Outcome Measures: Unilateral knee flexion and side-lying hip adduction maximum voluntary isometric contractions were monitored routinely in-season, two days post-match (≥40 hours [h]) during a three-season intervention period. Strength reductions greater than the tests' minimal detectable change percentage prompted intervention. HSI and time-loss groin injury burdens were calculated per 1000 player hours and compared with those from an immediately preceding two-season control period, to assess the effectiveness of the intervention.

Results: Across the intervention period, there was a decrease in HSI (4.98 days absence/1000h (19.8%) decrease) and time-loss groin injury burdens (0.57 days absence/1000h (49.1%) decrease) when compared with the control period.

Conclusions: A systems-based secondary injury prevention intervention shows preliminary positive findings in reducing HSI and time-loss groin injury burdens within a male professional football club, compared with usual care only.

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Subsequent musculoskeletal injury after concussion in National Football League players. British Journal of Sports Medicine, Ahead of Print: 1-7, 2024.

Objective: To assess whether National Football League (NFL) players diagnosed with a concussion have an increased risk of injury after return to football.

Methods: A retrospective cohort study analysed the hazard of subsequent time-loss lower extremity (LEX) or any musculoskeletal injury among NFL players diagnosed with a concussion in 2015-2021 preseason or regular season games compared with: (1) all non-concussed players participating in the same game and (2) players with time-loss upper extremity injury. Cox proportional hazards models were adjusted for number of injuries and concussions in the prior year, player tenure and roster position. Additional models accounted for time lost from participation after concussion.

Results: There was no statistical difference in the hazards of LEX injury or any musculoskeletal injury among concussed players compared with non-concussed players, though concussed players had a slightly elevated hazard of injury (LEX injury: HR=1.12, 95% CI 0.90 to 1.41; any musculoskeletal injury: HR=1.08, 95% CI: 0.89 to 1.31). When comparing to players with upper extremity injuries, the hazard of injury for concussed players was not statistically different, though HRs suggested a lower injury risk among concussed players (LEX injury: HR=0.78, 95% CI: 0.60 to 1.02; any musculoskeletal injury: HR=0.82, 95% CI: 0.65 to 1.04).

Conclusion: We found no statistical difference in the risk of subsequent injury among NFL players returning from concussion compared with non-concussed players in the same game or players returning from upper extremity injury. These results suggest deconditioning or other factors associated with lost participation time may explain subsequent injury risk in concussed players observed in some settings after return to play.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ There are contrasting reports regarding the risk of subsequent musculoskeletal injury following concussion, with few studies having robust data on potential confounders (eg, prior injury history, time at risk), detailed follow-up, or a comparative group of athletes without concussion but with time loss from sport.

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WHAT THIS STUDY ADDS

⇒ This study identified time lost due to injury, and not necessarily concussion, as a potential driver of risk for subsequent musculoskeletal injury after return to participation among professional American football players.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Clinicians should consider the increased risk of subsequent injury from deconditioning due to time away from sport, regardless of the cause of the time loss. These findings suggest that concussed athletes as well as those returning from an upper extremity injury may benefit from more robust conditioning prior to reintroduction to sport; the impact of such an approach on all injured athletes should be further investigated.

The effect of fresh and used ankle taping on lower limb biomechanics in sports specific movements. Journal of Science and Medicine in Sport, Ahead of Print: 1-7, 2024.

Objectives: We aimed to investigate the effects of ankle taping on lower extremity biomechanics related to injury development and how these effects change after sports-specific use.

Design: Randomized, repeated measures design with three conditions: Barefoot, tape applied fresh, and tape after sports-specific use (between-subject factor: sex).

Methods: Twenty-five healthy participants (ten female) performed sports-specific movements, including running, drop jumping, and 180° change of direction, under the three conditions. Kinetic and kinematic data were collected using 3D motion capturing and force platforms.

Results: Tape applied fresh and tape after sports-specific use significantly reduced peak ankle inversion. Biomechanical risk factors for anterior cruciate ligament or running overuse injuries were either unchanged or decreased with tape applied fresh, except for the peak loading rate of the resultant ground reaction force, which increased between 4% and 18% between movement types. After 15 minutes of sports-specific use of the tape, the alterations induced by tape applied fresh remained for

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some biomechanical risk factors while they became closer to barefoot again for others, indicating a differential response to prolonged use of taping for different biomechanical variables.

Conclusions: Ankle taping protects the ankle joint by reducing biomechanical risk factors associated with ankle sprains, and most biomechanical risk factors for anterior cruciate ligament or running overuse injuries are not increased. Further research is needed to explore the duration of protective effects, variations across sports, and its impact on patients with chronic ankle instability, contributing to a more comprehensive understanding of ankle taping's influence on lower extremity biomechanics.

Visuomotor strategies for collision avoidance in athletes: Insights from virtual reality. Journal of Science and Medicine in Sport, Ahead of Print: 1-4, 2024.

Abstract - This study explores visuomotor control in athletes for collision avoidance using virtual reality. Thirty-nine athletes navigated dynamic scenarios, pursuing a virtual target while avoiding two to five virtual defenders. Confirmatory Factor Analysis validated a model that captured two features of scanning behavior based on head movements recorded during activity: the overall amount and temporal pattern. Linear mixed models showed that these features significantly differentiated successful from unsuccessful defender avoidance ($p < .05$), suggesting that efficient environmental scanning is crucial for collision avoidance while highlighting the potential of visuomotor interventions to reduce collision-related sport injuries.

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Pro Football Sports Science Research Update

What are the protocols and resources for sport-related concussion among top NCAA football programs? A cross-sectional survey of A5 schools. Clinical Journal of Sport Medicine, Ahead of Print: 1-7, 2024.

Objective: This study summarizes findings from a cross-sectional survey conducted among National Collegiate Athletic Association (NCAA) Division 1 football programs, focusing on sport-related concussion (SRC) protocols for the 2018 season.

Design: Cross-sectional survey study.

Setting: 65 football programs within the Autonomy Five (A5) NCAA conferences.

Participants: Athletic trainers and team physicians who attended a football safety meeting at the NCAA offices June 17 to 18, 2019, representing their respective institutions.

Intervention: Electronic surveys were distributed on June 14, 2019, before the football safety meeting.

Main outcome measures: Results for 16 unique questions involving SRC protocols and resources were summarized and evaluated.

Results: The survey garnered responses from 46 of 65 programs (response rate = 71%). For baseline testing, 98% measured baseline postural stability and balance, 87% used baseline neurocognitive testing, while only 61% assessed baseline vestibular and/or ocular function. Regarding concussion prevention, 51% did not recommend additional measures, while 4% and 24% recommended cervical compression collars and omega-3 supplementation, respectively. In postconcussion treatment, 26% initiated aerobic exercise 1 day postconcussion if symptoms were stable, 24% waited at least 48 hours, 4% waited for the athlete to return to baseline, 11% waited until the athlete became asymptomatic, and 35% determined procedures on a case-by-case basis.

Conclusions: Most institutions assessed postural stability/balance and neurocognitive functioning at baseline and introduced light aerobic exercise within 48 h postconcussion. There was variation in baseline assessment methods and concussion prevention recommendations. These survey findings deepen our understanding of diverse SRC protocols in NCAA football programs.

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